

- Q.27 Explain the manufacturing of steel by electric furnaces.
- Q.28 Explain T-T-T diagram.
- Q.29 What is pyrometer? Name its different types.
- Q.30 Explain gas Carburizing.
- Q.31 Explain lead bath furnace.
- Q.32 Write down the process of austenite formation.
- Q.33 Write the properties of hinalium.
- Q.34 What are quenching cracks defect.
- Q.35 Explain flame hardening and induction hardening.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Describe Iron-Carbon Equilibrium Diagram in Detail. Explain various stages.
- Q.37 Explain the manufacturing of pig iron with a neat sketch of blast furnace.
- Q.38 Explain various defects during heat treatment. Also explain their causes and prevention.

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5th Sem / T&D, Found. & Forg. Subject:- Heat Treatment

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The material obtained from blast furnace by the reduction of iron ore is
- a) Wrought iron b) Pig iron
- c) Cast iron d) Carbon steel
- Q.2 Which of the following is not an ore of iron?
- a) Hematite b) Bauxite
- c) Limonite d) Siderite
- Q.3 Bronzes are the alloys of
- a) Copper-tin b) Copper-zinc
- c) Copper-nickel d) Copper-nickel-zinc
- Q.4 Utensils are generally made of
- a) Duralumin b) Hinalium
- c) Magnelium d) None of the above
- Q.5 Silicon improves the _____ properties of steel.
- a) Electrical b) Mechanical
- c) Thermal d) Magnetic

- Q.6 The Time-Temperature -Transformation (TTT) diagram is also known as _____
- a) T-Curve b) S-Curve
c) B-Curve d) None of the above
- Q.7 In annealing process, cooling is done in
- a) Water b) Air
c) Oil d) Furnace
- Q.8 Which of the following is used for making the resistance of heating elements?
- a) Nichrome b) Invar
c) Phosphor bronze d) Manganin
- Q.9 German silver contains
- a) 1% silver b) 2% silver
c) 3% silver d) No silver
- Q.10 Ammonia gas is introduced on steel surface, the process is
- a) Nitriding b) Cyaniding
c) Carburizing d) Ammonizing

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Stainless steel has _____ thermal conductivity than carbon steel.

- Q.12 Pig iron is the basic material from which the other forms such as wrought iron and steels are manufactured. (True/False)
- Q.13 Open hearth process is also known as Siemens process. (True/False)
- Q.14 Cobalt causes carburization of steel. (True/False)
- Q.15 Hematite contains about _____ % Iron.
- Q.16 Melting point of Iron starts above the temperature _____ in degree Celsius.
- Q.17 _____ is an alloy of copper, tin and zinc.
- Q.18 The addition of _____ in steel lowers the toughness and transverse ductility.
- Q.19 18-4-1 high speed steel contains _____ % Tungsten
- Q.20 In cupola furnace _____ coal is used.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Name different types of cast iron.
- Q.22 Write the use of dead mild steel.
- Q.23 Write down about different copper alloys.
- Q.24 What are characteristics of stainless steel?
- Q.25 Explain any two special steels.
- Q.26 What are different allotropic forms of iron? Explain.